

Library Management System - Test Cases

Test Case 1: Borrowing With Basic Membership

Login as Basic member
Borrow 2 books successfully
Try borrowing 3rd book

Expected: Borrowing limit exceeded

Test Case 2: Borrowing With Premium Membership

Login as Premium member
Borrow 5 books successfully
Try borrowing 6th book

Expected: Borrowing limit exceeded

Test Case 3: Late Return Fine

Update due date manually in database
Return borrowed book

Expected: Fine calculated based on delayed days

Test Case 4: 25% Damage Fine

Borrow a book
Return book with 25% increased damage
Expected: ₹100 fine

Test Case 5: 50% Damage Fine

Return book with 50% increased damage
Expected: ₹300 fine

Test Case 6: Fully Damaged Book

Return book with 100% damage
Expected: Full book price fine and status changed to Lost

Test Case 7: Duplicate Borrowing Validation

Borrow a book
Try borrowing same book again
Expected: Duplicate borrowing exception

Test Case 8: Unavailable Book Validation

Try borrowing unavailable book
Expected: No available copy found

Test Case 9: Stored Function Execution

Return a borrowed book

Expected: PostgreSQL function updates borrowing, fines, and book copy

Test Case 10: Available Books Report

Open reports menu

View available books by category

Expected: Books grouped category-wise

Test Case 11: Overdue Books Report

Create overdue borrowing

Open overdue books report

Expected: Delayed days displayed

Test Case 12: Pending Fines Report

Open pending fines report

Expected: Members with pending fines shown

Test Case 13: Most Borrowed Books Report

Open most borrowed books report

Expected: Books ordered by borrow count

Test Case 14: Member Borrowing Summary

Enter member id

Expected: Borrowed books, returned books, and pending fines displayed

Test Case 15: Fine Payment

View pending fines

Pay fine

Expected: Fine marked as paid

Test Case 16: Search Books

Search by title, author, or category

Expected: Matching books displayed

Test Case 17: Search Members

Search by partial email or phone number

Expected: Matching members displayed